

Folding Divider for Limiting Distractions and Increasing the Span of Useful Study Area

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Criterion A: Analysis of Design Opportunity

i. Identified Design Problem

As a full-time student taking a schedule full of college-level courses, I've found effective personal study time to be one of the most critical factors in achieving academic success. However, in shared or busy environments like classrooms, libraries, or even home study spaces, distractions often make it difficult to focus. Many students, including myself, struggle to find a sense of personal space in these open areas. With the intention of solving this problem in mind, I began researching potential product designs that could increase efficiency and comfortability when studying.

Initially, I surveyed a group of students enrolled in at least one AP/IB course to find out just how problematic distractions are among students enrolled in rigorous courses. All students surveyed attend an average public school in the midwestern United States, making them an ideal focus group for my design solution. The results of my survey can be seen below.

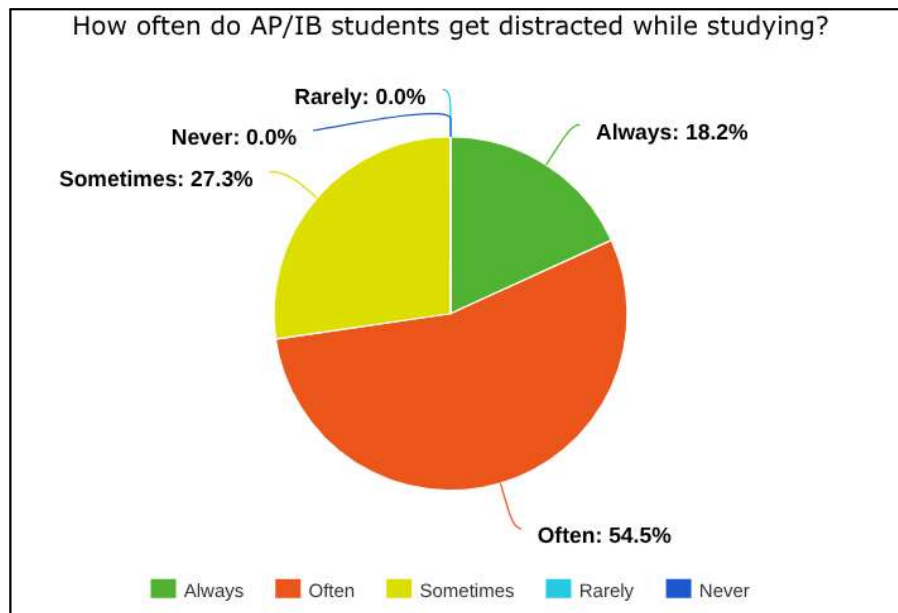


Figure 1. Percentage of Students Reported Being Distracted

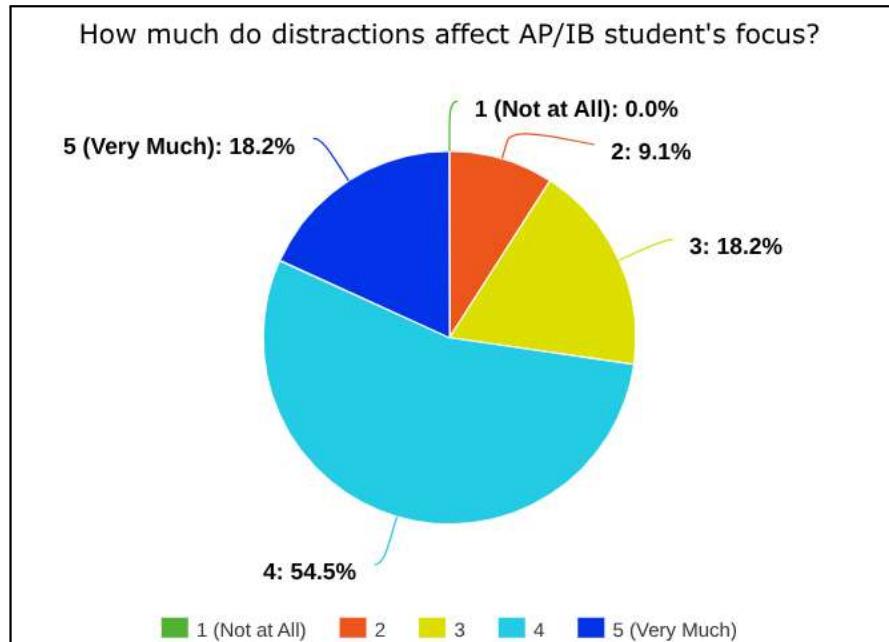


Figure 2. How Much Distractions Affected Focus (Scale 1-5)

72.7% of sampled high school students enrolled in college-level classes reported being often or always distracted while studying (Fig. 1). This aligns with responses from the 1-5 focus scale in my survey (Fig. 2), where most students selected values in the upper range. These higher ratings show that distractions don't just happen occasionally; they noticeably interfere with students' ability to stay on task. Together, both data sets point to a consistent pattern: students in demanding courses are struggling to maintain focus, reinforcing the need for a solution that helps reduce these interruptions.

This observed local trend can also be seen nationwide. The National Center for Education Statistics reported in 2024 that about one in four public school leaders identified student inattention as having a severe negative impact on learning (NCES, 2024). This growing problem underscores the need for tools that help students create focused, distraction-free study zones, even in shared spaces.



Figure 3. Image Showing Distraction-Riddled Study Area
Source: The Scratching Post Student Publication (2017)

After confirming that distractions significantly interfere with students' study time and that the problem is widespread, I began developing a product concept aimed at addressing these challenges. My goal was to identify a design that aligns with student needs and effectively reduces the distractions that limit their ability to focus. To determine the initial guidelines for my design, I surveyed the same group of AP and IB students as before (see Figure 1, Figure 2.).

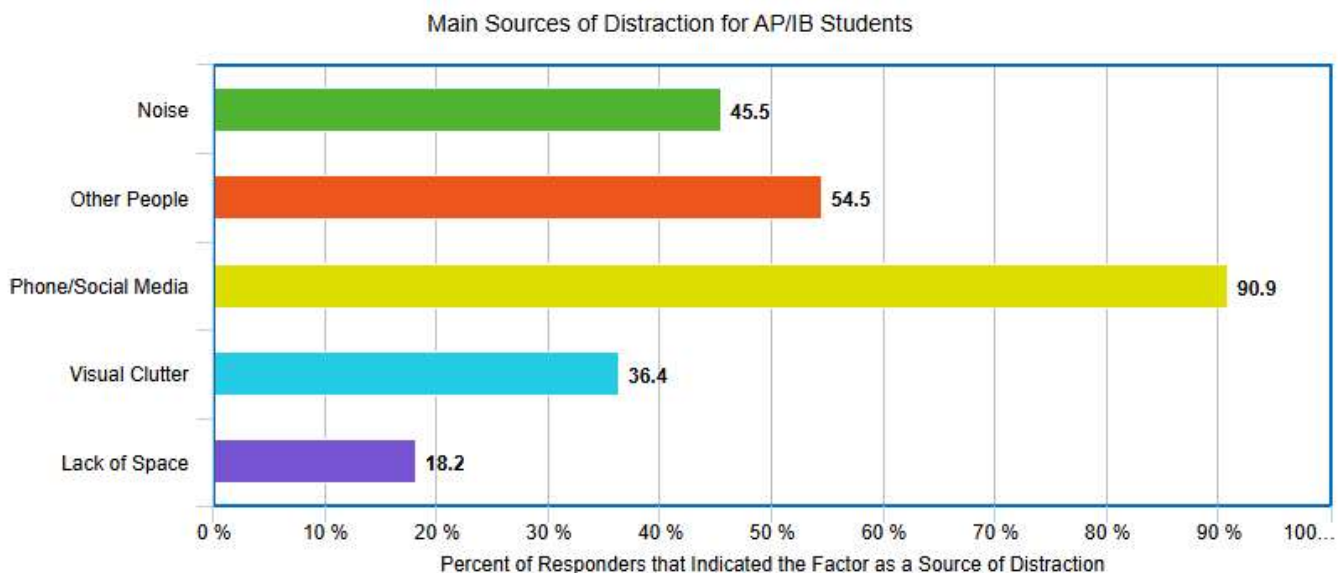


Figure 4. Main Sources of Distraction

The survey showed that phones and social media were the biggest distractions (Fig 4.). After thinking it through, I realized those habits had more to do with personal discipline than the study space itself. That shifted my focus toward the next major issues students mentioned: noise, visual clutter, and the presence of other people. With those very issues in mind, I looked at the market to understand what solutions already existed and to figure out how I could approach the problem in a way that actually met students' needs.



Figure 5. A Student Distracted by Her Peers

Source: Learning LiftOff

Market Analysis:

(Prices and Images from [Amazon.com](https://www.amazon.com) and [Walmart.com](https://www.walmart.com))

Product 1

Product Name: ReFocus Raw Freestanding Acoustic Desk Divider	Price: \$199.00 USD
Product Pros & Cons of Design as Related to Design Problem	
Pros • Cuts down on background noise • Creates a smaller, defined study zone • Requires no technology or apps (works passively)	Cons • Much too expensive for most students • Large and difficult to move • Reduces noise and visual distractions from only one side
Product Image: 	

Product 2

Product Name: Bouncyband Portable Desktop Privacy Partition	Price: \$29.33 USD
Product Pros & Cons of Design as Related to Design Problem	
Pros • Lightweight and very easy to carry • Blocks most visual distractions in shared spaces • Inexpensive compared to larger panels	Cons • Limited noise reduction • Less stable and durable than heavier panels • Takes up desk space and can feel cramped for some users
Product Image:	

Product Name: Bouncyband Portable Desktop Privacy Partition	Price: \$29.33 USD
	

Product 3

Product Name: SoundSorb™ Desktop Privacy Panels	Price: \$29.99 USD
Product Pros & Cons of Design as Related to Design Problem	
Pros • Strong noise-dampening performance • Clips onto a desk without taking much workspace • More affordable than full acoustic dividers	Cons • Only reduces noise, does not block all visual distractions • Fixed size limits coverage • Not ideal for students who frequently change study locations
Product Image: 	

ii. Design Brief

Maintaining focus in shared or busy study environments is a consistent challenge for students enrolled in demanding courses. Survey data from AP and IB students shows that most encounter frequent distractions, and many report that these interruptions significantly affect their ability to stay engaged. Because these students form the primary target audience, understanding the specific sources and impact of their distractions is central to defining the problem. National data reflects the same pattern, with many schools reporting that inattention is increasingly affecting learning outcomes (NCES, 2024). Taken together, the evidence shows a strong need for a product that helps students establish personal study space, limit surrounding disruptions, and better sustain concentration. The issue is widespread and directly tied to academic performance, reinforcing the importance of an effective design solution.

The research plan incorporates both primary and secondary sources to guide development. Primary research includes student surveys that identify distraction frequency and types and observations of how students interact with shared spaces. Secondary research involves reviewing studies on attention and learning, analyzing existing products aimed at reducing distractions, and examining national reports on barriers to student concentration. Product analysis so far indicates that portability, noise reduction, and the ability to limit visual or spatial distractions are the most key considerations. These findings will shape a design that provides students with a compact, functional tool that creates a more controlled study environment. The intended outcome is a solution that supports sustained focus, integrates smoothly into daily academic routines, and directly addresses the needs outlined by consumers.

iii. Design Specification

Specification	Requirement	Justification	Testing Method	Priority
1. Aesthetics	1.1 Should have a clean, modern look	1.1 A visually calm product helps reduce sensory overload and keeps the user focused during studying. A modern style increases acceptance	-User preference survey -Color analysis	Low

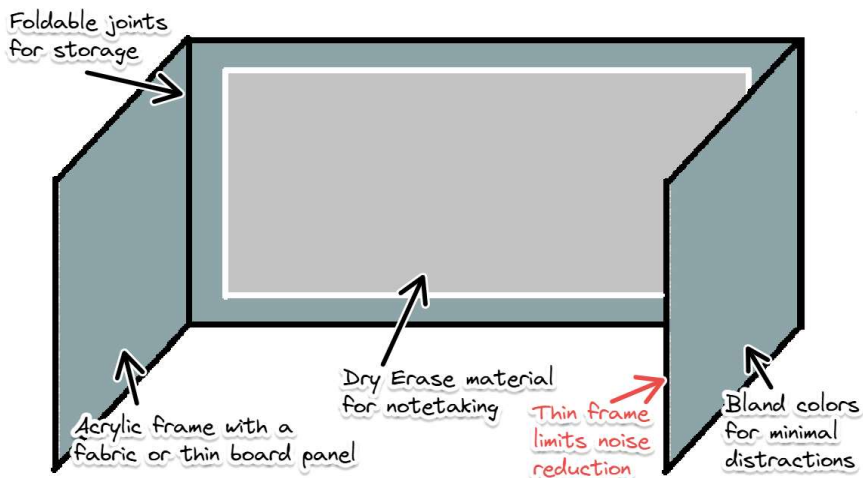
	1.2 Color scheme should be neutral or muted	among students and fits in with typical school or home study spaces. 1.2 Strong colors can be distracting. Neutral tones blend easily into any environment, making the product more flexible for various study locations.		
2. Function	2.1 Must reduce visual distractions 2.2 Should reduce ambient noise levels 2.3 Must remain stable during use	2.1 The core purpose of the product is to shield the user from surrounding movement so they can stay focused 2.2 Noise is one of the most common study issues reported, so improving acoustic comfort directly supports the design problem. 2.3 A product that shifts or wobbles becomes distracting or unusable, undermining its purpose.	- Target audience survey -Usability testing	High
3. Target Audience	3.1 Should appeal to students 3.2 Must be usable in user's common study environments	3.1 Students are the primary users, so the design must match their preferences for both looks and use. 3.2 Students switch between libraries, bedrooms, cafés, and classrooms. The product must adapt well to all.	- Target audience survey -Testing in real environments	High

4. Production Constraints	4.1 Must be manufacturable using available workshop tools 4.2 Materials and construction must stay within budget	4.1 Ensures the design can actually be built within school facilities and IA limitations. 4.2 Keeps the project feasible and avoids expensive or unrealistic components.	-Tool availability check -Cost Analysis	Low
5. Material Selection	5.1 Must be lightweight but durable 5.2 Should use materials with good sound absorption	5.1 Students carry study tools around, so the product needs to be easy to move while still lasting through daily use. 5.2 If noise reduction is part of the design goal, the materials must reinforce that function.	-Weight checks and durability tests -Acoustic testing	High
6. Size	6.1 Must fit on a standard desk with room for study materials 6.2 Should be compact enough for easy storage or transport	6.1 Ensures the product can be used everywhere students typically work without limiting usable space. 6.2 Students often pack up quickly or move between locations, so the product can't be bulky.	-Fit tests on various desk -Backpack/locker storage tests	Medium
7. Quality	7.1 Must withstand frequent handling and repositioning	7.1 Regular use causes wear, so good build quality keeps the product reliable over time.	-Repetitive-use stress testing	Medium

	7.2 All edges and surfaces should be smooth and safe	7.2 Prevents irritation or injury and keeps the product comfortable to use.	- Physical inspection	
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Criterion B: Conceptual Design

i. Initial Sketches

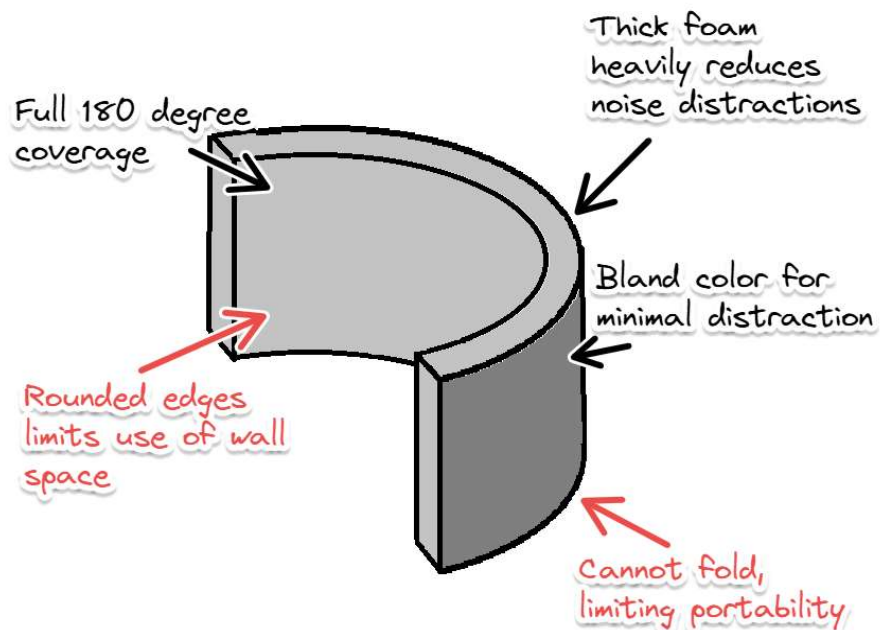


Concept #1

The **Folding Desk Blind** is a simple panel that sits on top of the desk and can fold flat for storage when not in use. It blocks visual distractions from both sides of the desk and is lightweight enough to move between study locations easily.

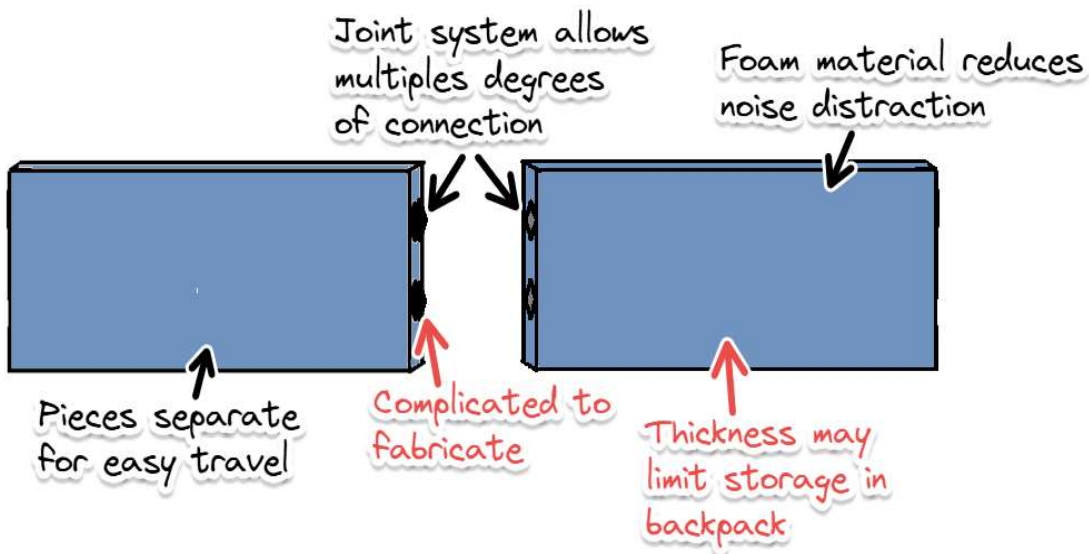
Concept #2

The **Semi-Circle Noise-Reduction Barrier** is a curved foam panel that wraps around the desk, blocking visual distractions and reducing ambient noise. Its shape allows the user to reach the desk easily while providing a semi-enclosed, focused study space. The design adaptable to different desks, making it practical for shared study environments.



Concept #3

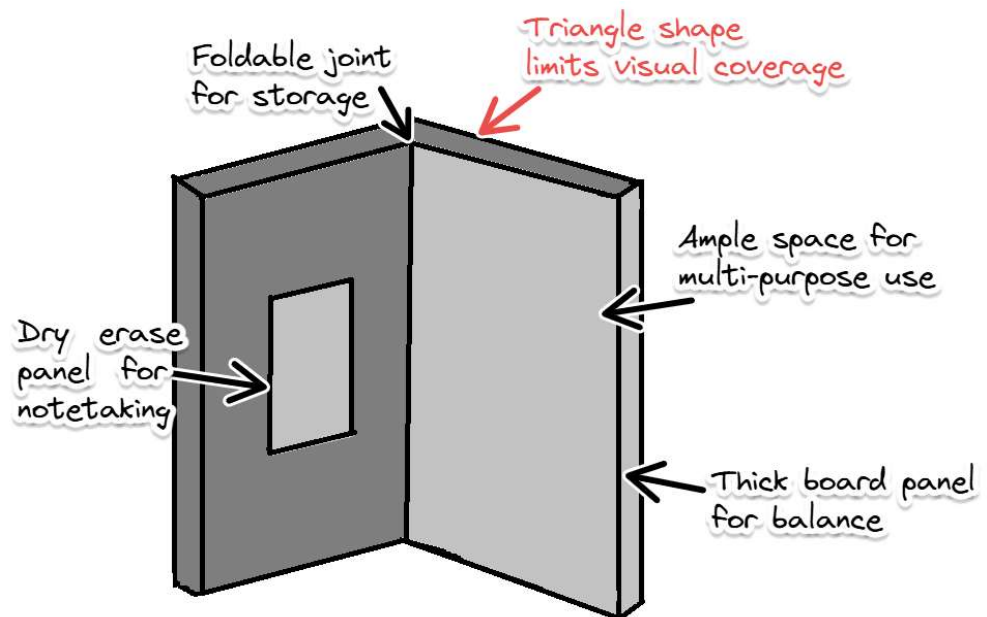
The **Adjustable Jointed Barrier** uses connected panels with a simple joint system that lets the user change the shape and angle of the barrier. It can shift from a straight line to a partial curve, allowing different levels of



coverage depending on the study environment. The panels block visual distractions and create a more controlled study space while staying easy to fold and carry.

Concept #4

This **Two-Panel Divider** folds at a single hinge and opens into a chosen angle that blocks outside views. One side includes a dry erase area for quick notes, and the other provides open space for whatever task you're working on. The panels are thick enough to stand steady on a desk without extra support.



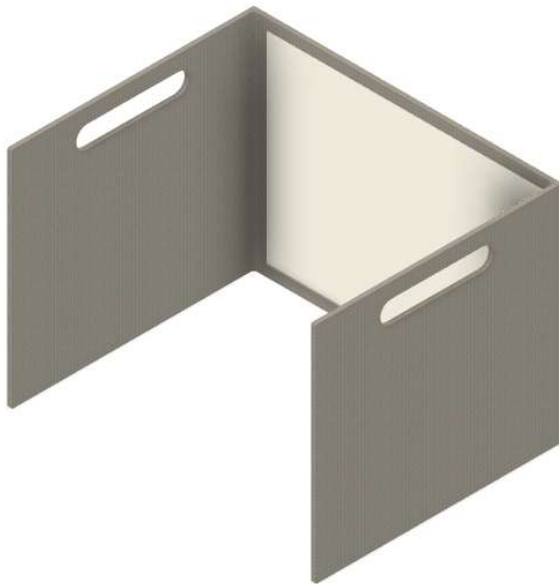
ii. Developed Designs

Based on a logistical evaluation of my designs and the consideration of material, time constraints, and peer approval, I determined that I would develop two of my initial

designs: Concept #1 (Tri-Panel Folding Desk Blind with Dry Erase Surface) and Concept #2 (Semi-Circle Noise-Reduction Barrier). Concept #1 appealed to me because of its multifunctionality and compact design, while Concept #2 offered a practical solution for improving focus through the reduction of noise distractions. To get a better idea of which concept to proceed with, I modelled both concepts in the 3D CAD software Autodesk Fusion 360.

Concept #1 (CAD Model)

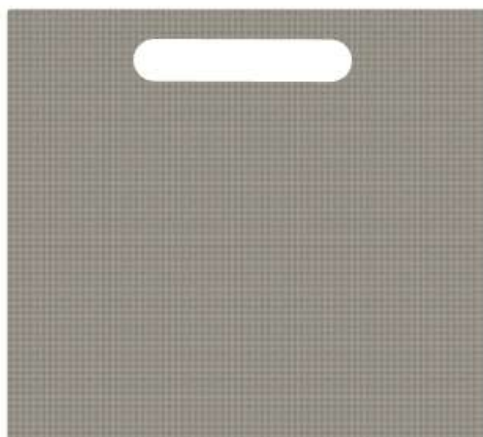
Material (tentative): Acrylic sheet with cloth covering, laminated hardboard.



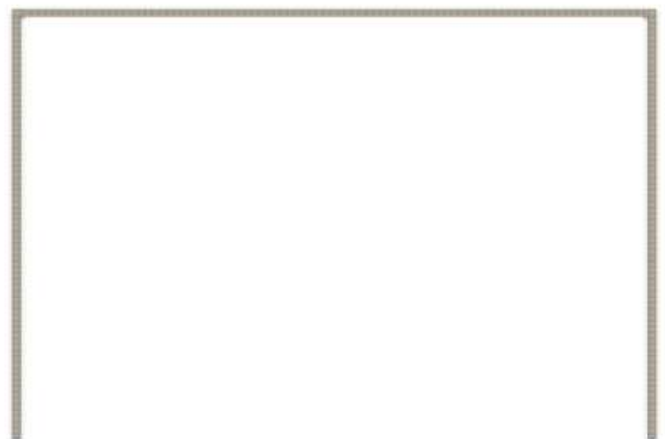
Home View



Front View



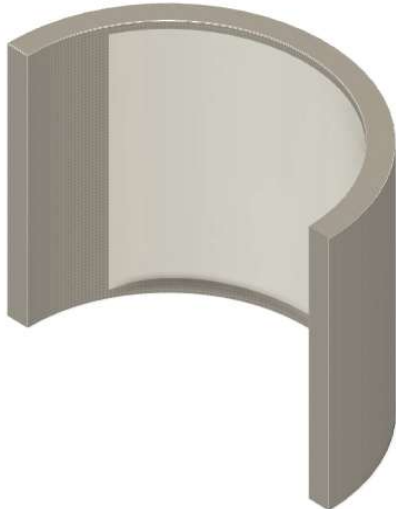
Side View



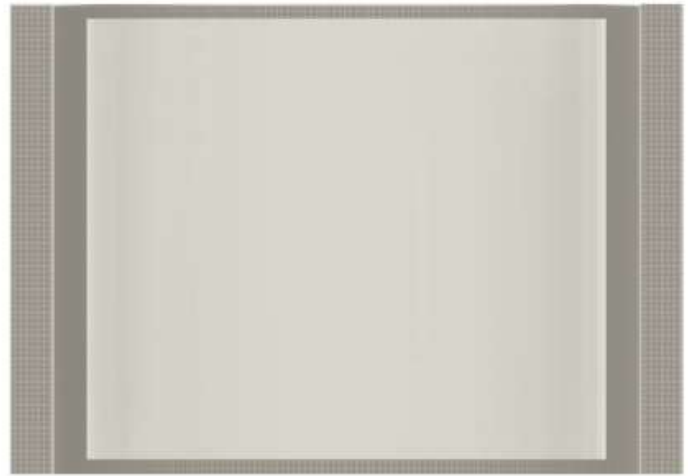
Top View

Concept #2 (CAD Model)

Material (tentative): Cloth covered foam board, laminated hard board.



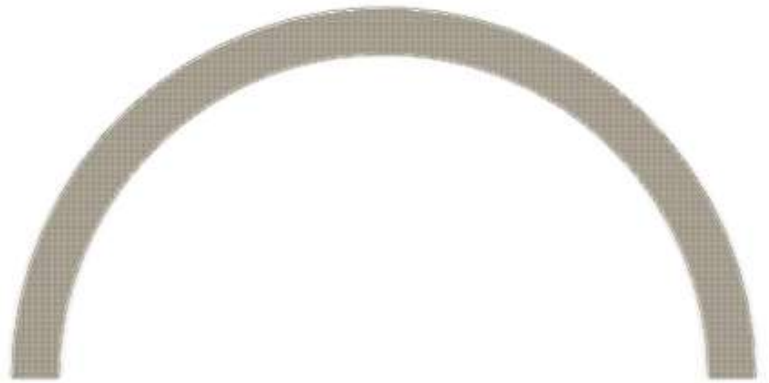
Home View



Front View



Side View



Top View

iii. Final Design Justification

Now that I had two analyzable 3D CAD models, the next step was to determine my final design choice. I did this by scoring each design against the specification rubric in the Criterion A section. Every section on the rubric is given a score from 1-10 based on the designs performance in each category. The results have been compiled into a table below:

Concept #1 (Tri-Panel Folding Desk Blind with Dry Erase Surface)

Score is based on testing methods outlined in the “testing method” section of the specification table. 1 being ‘neglected completely’, 5 being ‘addressed perfectly’.

Scores for untestable aspects are based on the ideal design concept.

Specification (graded on scale 1-5)	Score
Aesthetics - 1.1 Should have a clean, modern look	5
Aesthetics - 1.2 Color scheme should be neutral or muted	5
Function - 2.1 Must reduce visual distractions	5
Function - 2.2 Should reduce ambient noise levels	4
Function - 2.3 Must remain stable during use	5
Target Audience - 3.1 Should appeal to students	4
Target Audience - 3.2 Must be usable in user’s study environment	5
Production Constraints - 4.1 Must be manufacturable using available workshop tools	5
Production Constraints - 4.2 Materials and construction must stay within budget	4
Material Selection - 5.1 Must be lightweight but durable	4
Material Selection - 5.2 Should use materials with good sound absorption	3
Size - 6.1 Must fit on a standard desk with room for study materials	5
Size - 6.2 Should be compact enough for easy storage or transport	5
Quality - 7.1 Must withstand frequent handling and repositioning	4
Total Projected Score	63/70

Concept #2 (Semi-Circle Noise-Reduction Barrier)

Score is based on testing methods outlined in the “testing method” section of the specification table. 1 being ‘neglected completely’, 5 being ‘addressed perfectly’.

Scores for untestable aspects are based on assumption and the design concept.

Specification (graded on scale 1-5)	Score
Aesthetics - 1.1 Should have a clean, modern look	3
Aesthetics - 1.2 Color scheme should be neutral or muted	5
Function - 2.1 Must reduce visual distractions	5
Function - 2.2 Should reduce ambient noise levels	5
Function - 2.3 Must remain stable during use	5
Target Audience - 3.1 Should appeal to students	3
Target Audience - 3.2 Must be usable in user’s study environment	4
Production Constraints - 4.1 Must be manufacturable using available workshop tools	3
Production Constraints - 4.2 Materials and construction must stay within budget	4
Material Selection - 5.1 Must be lightweight but durable	5
Material Selection - 5.2 Should use materials with good sound absorption	5
Size - 6.1 Must fit on a standard desk with room for study materials	5
Size - 6.2 Should be compact enough for easy storage or transport	2
Quality - 7.1 Must withstand frequent handling and repositioning	5
Total Projected Score	59/70

Summary of Results:

Concept 1, the tri-panel folding desk blind with a dry erase surface, achieved a higher overall score of 63 out of 70. This design performed strongly across most criteria, particularly in aesthetics, stability, size, and ease of manufacture. The folding panel layout effectively reduces visual distractions while remaining compact enough for storage and transport. Its stable form and neutral appearance also make it well suited for a typical student study environment. The added dry erase surface increases functionality without adding significant cost or complexity.

Concept 2, the semi-circle noise-reduction barrier, scored 59 out of 70. While it performed very well in sound absorption and durability, its curved shape makes it bulkier and less practical to store or transport. It also scored lower in manufacturability, as producing a precise curved form would be more challenging with the available workshop tools. Additionally, the design is less flexible for different study setups.

Overall, Concept 1 offers a more balanced solution to both visual and auditory distractions. Although Concept 2 provides better noise reduction, Concept 1 better meets the full range of design requirements. For these reasons, Concept 1 was selected as the final design and will be taken forward for further development.

Criterion C: Development of a Detailed Design

i. Prototype Development Process Justification

The prototype will be created using a planned process that focuses on functionality, practicality, and ease of manufacture within a school workshop environment. All materials and methods will be chosen to allow realistic testing while keeping costs low and enabling easy modification.

The main structure will be made from corrugated cardboard. This material will be selected because it is lightweight, inexpensive, and easy to cut using basic tools. It will provide enough rigidity to keep the panels upright while still allowing them to fold,

making it suitable for testing the tri-panel design without committing to more expensive materials.

A dry erase surface will be added to the center panel using laminated white card. This will simulate the function of a real whiteboard and allow writing and erasing during testing. The laminated surface will be easy to attach, replace, or adjust as the design develops.

The folding mechanism between panels will be created using integrated cardboard hinges formed by scoring fold lines. This method will be chosen because it removes the need for separate hinges, keeping the prototype lightweight and simple. It will also allow panel angles to be adjusted easily during testing.

Assembly will be done using hot glue and double sided tape. Hot glue will be used for structural joints, while double sided tape will be used for components that may need repositioning. These methods are quick, accessible, and appropriate for a school workshop.

To test noise reduction, a thin layer of felt will be attached to the outer surfaces of the side panels. Felt will be chosen for its basic sound absorbing properties and ease of installation, allowing early evaluation of acoustic performance.

The prototype will be manufactured by cutting panels to size, scoring fold lines, attaching the dry erase surface, and assembling all components in a logical sequence. Overall, these materials and manufacturing methods will allow the prototype to be built efficiently, tested realistically, and refined based on feedback, while accurately representing the intended final design.

ii. Technical Design Plan

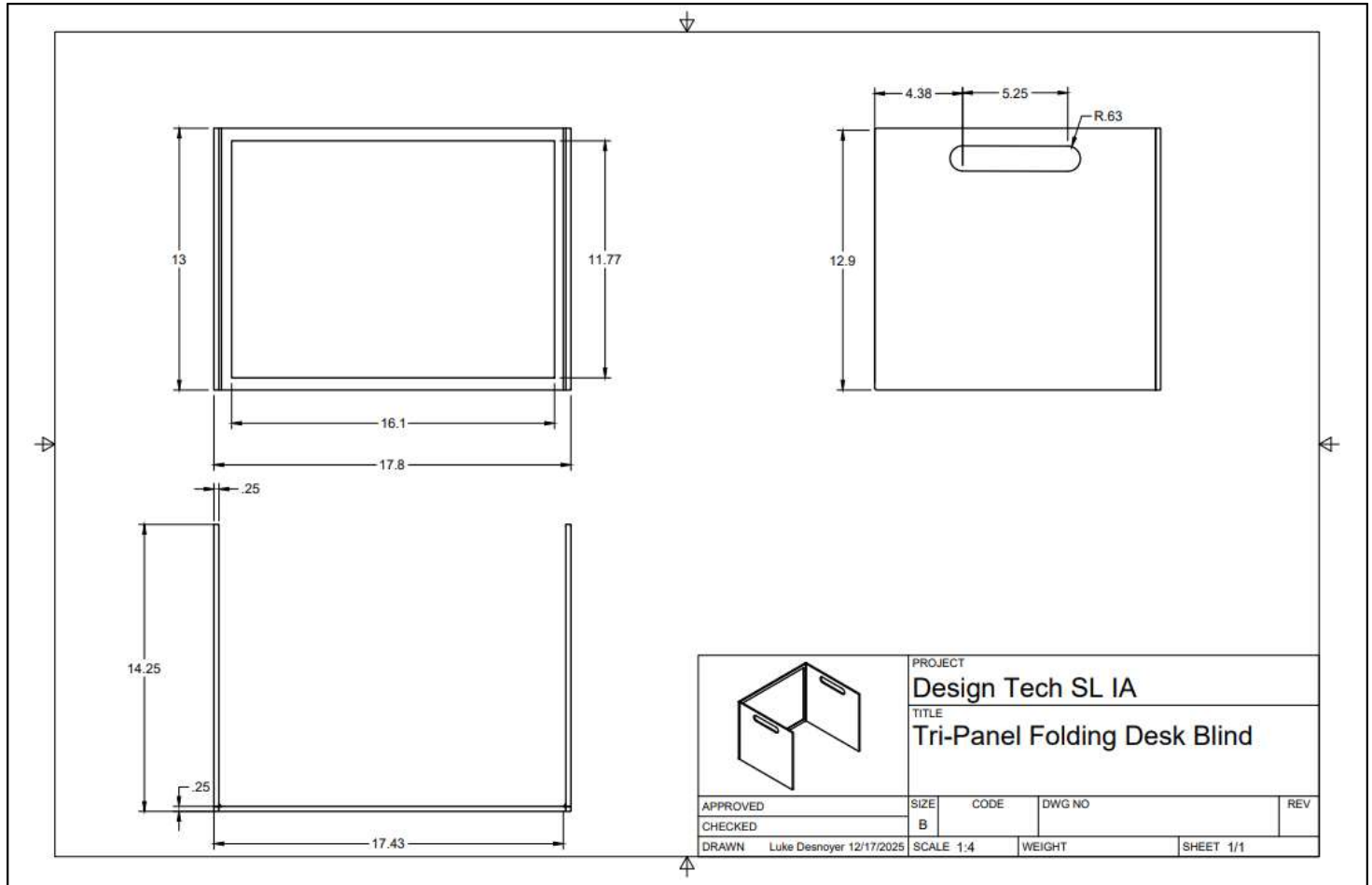


Figure 6. Design Sheet

Bill of Materials (Prototype):

Item No.	Component	Material	Quantity	Description and Purpose
1	Left Panel	Corrugated cardboard	1	Forms the left side of the structure, providing stability and partial enclosure for the user.
2	Center Panel	Corrugated cardboard	1	Acts as the main structural panel and

				central support, connecting the left and right panels.
3	Right Panel	Corrugated cardboard	1	Forms the right side of the structure, completing the enclosure and improving rigidity.
4	Integrated Cardboard Hinges	Scored cardboard fold lines	2	Allows the panels to fold inward and outward, enabling adjustability and easy storage.
5	Dry Erase Surface	Laminated white card	1	Provides a reusable writing surface for notes, reminders, or planning.
6	Adhesive (Structural)	Hot glue	As required	Permanently bonds cardboard components to ensure structural integrity during use.
7	Adhesive (Temporary)	Double sided tape	As required	Allows removable attachment of components for testing, adjustments, or prototyping changes.
8	Acoustic Layer	Felt sheet	3 pieces	Reduces sound reflection and absorbs noise within the enclosure to improve acoustic performance.

iii. Construction Plan

Step	Process	Equipment	Scheduling	Quality Control	Risk Assessment
1	Measure and mark cardboard panels	Ruler, pencil	Day 1	Verify dimensions against drawings	Minor cuts from tools
2	Cut panels to size	Craft knife, cutting mat	Day 1	Clean, straight edges	Blade injury risk
3	Score hinge fold lines	Ruler, blunt knife	Day 1	Consistent fold depth	Over-scoring weakens cardboard
4	Fold panels into shape	Hands, flat surface	Day 1	Smooth, aligned folds	Cardboard tearing
5	Attach panels structurally	Hot glue gun	Day 2	Strong, even joints	Burns from hot glue
6	Apply dry erase surface	Double sided tape	Day 2	Secure, flat adhesion	Misalignment during placement
7	Install acoustic felt layers	Hot glue, scissors	Day 2	Full coverage, no gaps	Glue burns, scissors cuts
8	Final inspection and testing	Visual check	Day 2	Stability and functionality	Structural failure if rushed

Criterion D: Testing and Evaluation

i. Testing Strategy Justification

The testing strategies outlined in the design specification will be applied to assess my prototype's performance as a marketable product. A high score indicates that the

criteria is acceptable for product release. A low score indicates that improvements must be made in the next design version.

Priority	Testing Technique	Justification
1	Usability testing	Determines whether users can correctly operate the divider without instruction, measuring setup time, error rate, and ease of folding. Directly affects functional effectiveness.
2	Strength testing	Evaluates the divider's ability to support its own weight, resist tipping, and endure repeated folding cycles without structural failure.
3	Sizing tests / measurements	Verifies that the divider meets design specifications, fits standard room dimensions, and folds within defined storage limits.
4	Acoustics testing	Measures sound reduction in decibels across the divider, providing quantitative evidence of noise dampening performance.
5	Target audience survey	Collects data on user requirements, perceived usefulness, and satisfaction to assess alignment with intended use cases.
6	Cost Analysis	Calculates material and manufacturing costs to determine economic feasibility and potential pricing constraints.
7	Tool availability check	Confirms that required tools and processes are accessible within the intended manufacturing environment.
8	Color Analysis	Assesses visual contrast and finish options but does not influence mechanical performance or core functionality.

ii. Design Specification Testing

Now that I had fabricated a prototype, I tested it against the design specification criteria to determine the overall feasibility of my product in the market. High school and college students are used as the sample for the target audience surveys.

1-5 scoring is based on the same scale as section Biii.

1. Aesthetics	1.1 Should have a clean, modern look
	1.2 Color scheme should be neutral or muted
Testing Strategy	Testing Results
Target audience survey	90% of users indicated that they approve of the design's aesthetics and would agree that the design is "modern."
Color analysis  Figure 7. Color Showcase	- All colors used in the final prototype (other than exposed cardboard) are muted. - Cardboard sticks out as cheap
Overall Score:	4/5

2. Function	2.1 Must reduce visual distractions
	2.2 Should reduce ambient noise levels
	2.3 Must remain stable during use
Testing Strategy	Testing Results

Target audience survey	(User score out of 5 for each category) ● 5/5 for reducing visual distractions ● 3/5 for ambient noise level reduction ● 4/5 for stability
Usability testing	● Visual testing clearly shows that distractions are blocked ● Testing indicated that ambient classroom noise intensity was reduced by 36.1% (decibels) when listened is surrounded by prototype ● Testing indicated that the prototype is able to freestand in several configurations
Overall Score:	4/5

3. Target Audience	3.1 Should appeal to students
	3.2 Must be usable in user's common study environments
Testing Strategy	Testing Results
Target audience survey	● 86% of users indicate that the prototype design appeals to them or is "promising"
Usability testing in school environment*	*Due to restrictions, I was unable to test in live classrooms. ● Design shop simulations indicate that the prototype would perform well in classroom environment
Overall Score:	4/5

4. Production Constraints	4.1 Must be manufacturable using available workshop tools
	4.2 Materials and construction must stay within budget
Testing Strategy	Testing Results
Tool availability check	All necessary tools outlined in construction plan are present in workshop
Cost Analysis	Total production cost of the prototype was \$0 (all material was recycled or provided by school)
Overall Score:	5/5

5. Material Selection	5.1 Must be lightweight but durable
	5.2 Should use materials with good sound absorption
Testing Strategy	Testing Results
Strength testing  Figure 8. Drop Testing	- Prototype can withstand fall from standard desk without major structural damage - Some damage is evident at falls from increased heights over 6 feet
Acoustic measurement	Felt material absorbs audio by up to 65%

Overall Score	3.5/5
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6. Size	6.1 Must fit on a standard desk with room for study materials
	6.2 Should be compact enough for easy storage or transport
Testing Strategy	Testing Results
Sizing tests/measurements	• Prototype fits on a standard school test with room to spare • Prototype fits in both a standard backpack and school locker
Overall Score	5/5

7. Quality	7.1 Must withstand frequent handling and repositioning
Testing Strategy	Testing Results
Strength tests   Figure 9. & Figure 10. Damage Sustained (7 foot drop)	• Prototype shows signs of wear after rough usage • Prototype looks cheap and quality would be questioned after damage is visible
Overall Score:	2.5/5

iii. Potential Improvements

The testing of my prototype indicated that my product was almost ready for the market; however, several aspects were still lacking, especially under the quality, material selection, and aesthetics criterion.

Improvement 1: Cardboard material swap

Criteria Addressed:	1.2 Color scheme should be neutral or muted
	5.1 Must be lightweight but durable
	7.1 Must withstand frequent handling and repositioning
Description: Testing indicates that the cheap cardboard material used to fabricate the prototype is not suitable for the final product because of aesthetic complaints and breakability. Cardboard was only used as a temporary material for prototyping, and was never intended to be the final material for the product. The material would hypothetically be changed to a laminated hardboard in a marketable design.	

Improvement 2: Sound absorption advancement

Criteria Addressed:	2.2 Should reduce ambient noise levels
Description: Whereas my prototype does reduce noise levels, auditory testing has indicated that this element can be improved to block even more sound. The cheap fabric used in my prototype could be swapped for synthetic suede that could reduce decibel levels by a more significant margin.	

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